

What does banbe cost to run?

A plain-English guide for Vietnam and the US | October 2026

The short answer. A launch version of banbe costs about **\$70 a month**. A busy one-city app (5,000 monthly users) costs about **\$130–180**. Even at 100,000 monthly users the servers cost about **\$1,300–1,700**, *if* text-message codes are bought from a cheap local provider. The photos are the cheapest part of the plan. The one decision that can multiply the bill is **who sends the phone-verification texts**.

1. Four stages of growth

A. Pilot	B. One city	C. Citywide	D. National
500 users	5,000 users	25,000 users	100,000 users
\$70	\$130–180	\$435–545	\$1,300–1,700
per month	per month	per month	per month
≈ 1.8 M VND	≈ 3.4–4.7 M VND	≈ 11–14 M VND	≈ 34–44 M VND

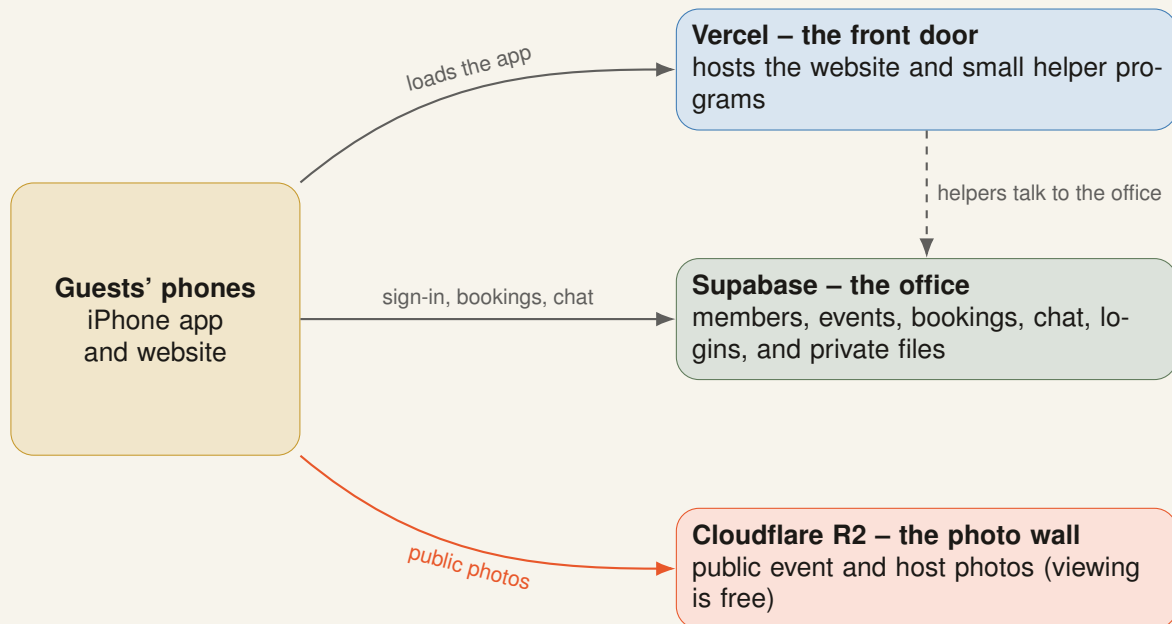
“Users” means *monthly active users*: different people who open the app at least once in a month. Prices in US dollars; 1 USD ≈ 26,100 VND. Taxes (VAT) are not included.



The two bars differ only in the price of the verification texts. Everything else is identical.

2. How banbe is built (the “hybrid” plan)

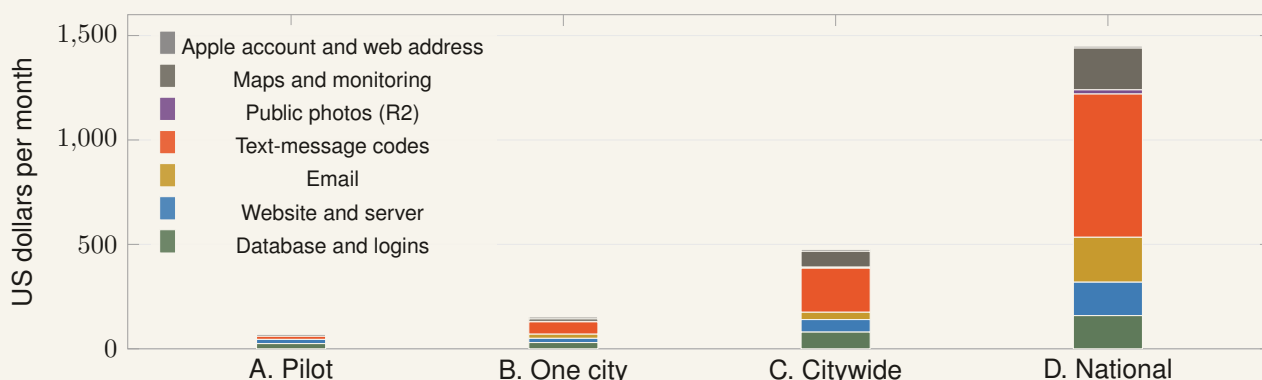
Think of banbe as a small venue with three rooms. We rent each room from the company that does it best and cheapest.



Room	What it does	Why this choice
Supabase	Remembers every member, event, booking and chat message; checks logins; keeps private files.	Already built in. Fixed monthly price (\$25) with a generous download allowance.
Cloudflare R2	Stores and shows the public photos.	Downloads are free . Photos are most of what people download, so this keeps the Supabase allowance safe.
Vercel	Hosts the website and small helper programs.	Already built in. \$20 a month for commercial use.

Why not use only Supabase? It works until photos are downloaded too often. That is what happened in October: the free allowance (5.5 GB) was used up and Supabase paused the app until 12 October. The hybrid plan prevents a repeat and costs almost nothing extra.

3. Where the money goes

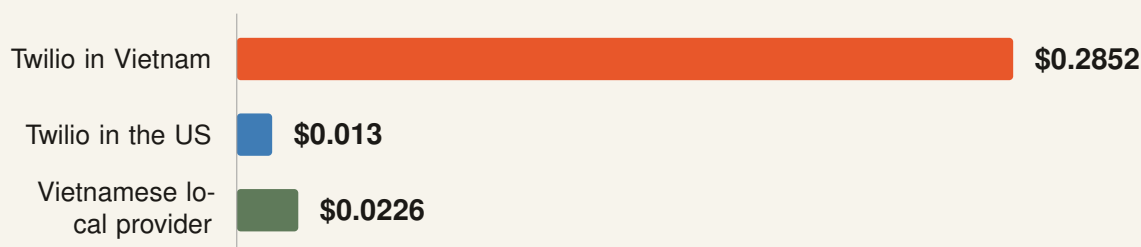


Stage B maps/monitoring uses the middle of a \$0–25 range. Totals: A \$69, B \$152, C \$476, D \$1,449.

Line	In plain words	Cost
Database and logins	Supabase Pro plan, plus a bigger “engine” as users grow	\$25 → about \$160
Website and server	Vercel Pro (required once banbe earns money)	\$20 → about \$160
Email	Tickets, receipts and verification emails (Resend)	free → about \$215
Text-message codes	A code texted to every new phone number	\$15 → \$686 (local) or \$86 → \$8,556 (Twilio)
Public photos	Cloudflare R2 storage; viewing is free	\$0 → about \$20
Apple account	Needed to publish the iPhone app and tickets in Apple Wallet	\$99 a year (\$8.25 a month)
Web address	For example <code>media.yourname.com</code>	about \$12 a year

4. The one decision that changes the bill: text-message codes

banbe asks every new member to verify a phone number. Each code is a text message, and the price per text depends heavily on the provider and the country.

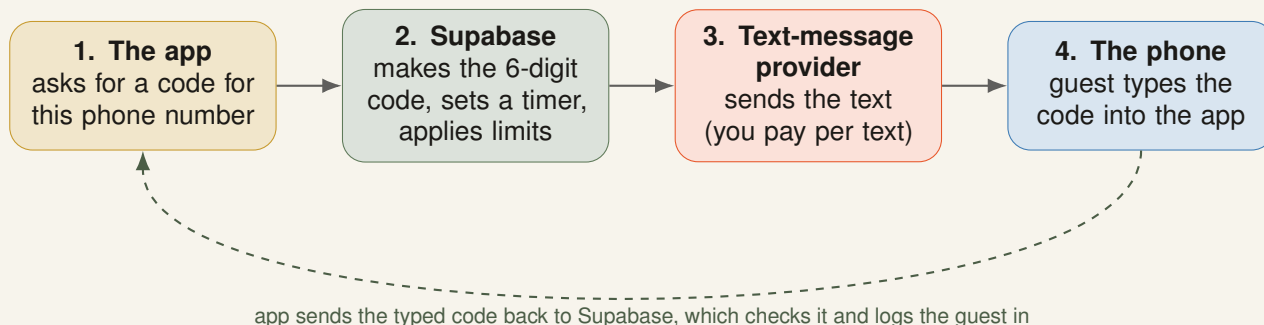


price of one text message, in US dollars (bars drawn to the same scale)

Example, one city (5,000 users, about 2,250 texts a month): a Vietnamese brandname provider ≈ **\$59**; Twilio in Vietnam ≈ **\$642**. The local price is SpeedSMS’s published list (590 VND per text plus a small monthly fee for the registered sender name); other local providers did not show prices, so ask for written quotes. Using a local provider needs a small piece of custom work in Supabase. In the US the standard provider is already cheap (about \$29 a month at this size).

5. Do we need a separate login service for phone codes?

No. Supabase already handles the whole login flow. It does not, however, send the text message itself: you must connect **one** text-message provider behind it. Supabase adds **\$0** on top of that provider's price, and there is no need for Firebase, Auth0 or Clerk.



Job	Who does it	Extra cost
Make, time-limit and check the code; apply limits; log the guest in	Supabase	included in the \$25 plan
Send the text message	A text-message provider you connect	price per text (section 6)
Email codes for sign-up and login	banbe's own server (Gmail today)	email service, section 3
Second-step login ("MFA") by text	Supabase add-on, not used by banbe	\$75 a month, avoid

How banbe uses it today. The phone check happens when a member verifies a phone number (web and iPhone). That is the step that triggers a text. The setting file in the project has the provider switched *off*, but that is only the developer's copy; **please check the live setting in the Supabase dashboard** (Authentication → Providers → Phone).

Providers. Supabase has Twilio, Twilio Verify, Bird (MessageBird), Vonage and Textlocal built in. Any other provider (a Vietnamese one, Telnyx, Plivo) connects through a "send SMS hook": a small piece of code that receives the code and hands it to the provider. The same hook can deliver by Zalo or email instead.

Built-in safety limits. Supabase allows **30 texts per hour for the whole project** by default, one code per person every 60 seconds, and 30 code-check attempts per 5 minutes per device. The 30-an-hour cap also limits fraud (at most about \$8.60 an hour on Twilio in Vietnam), but at about 9,000 texts a month (citywide stage) a busy evening could exceed it, so **raise it as the app grows**. Supabase advises adding a CAPTCHA too; check that your plan offers one.

Ways to need fewer texts (product choices, not priced here): verify the phone only when someone first books or hosts, not at sign-up; offer Zalo or email first in Vietnam with text as backup; skip re-verification on a returning device.

6. Text-message providers compared

Every provider below can send the verification code. They differ in price, in how much fraud protection comes included, and in how easily they plug into Supabase. Prices are per text message in US dollars.

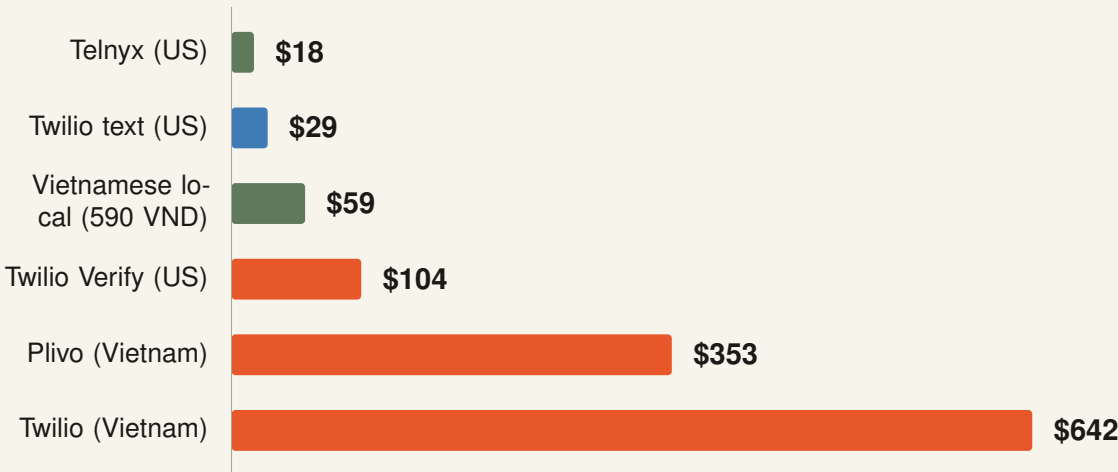
Vietnam (phone numbers starting +84)

Provider	Price per text	What to know
SpeedSMS (local)	\$0.017–0.030 (450–790 VND)	Published price list. Needs a registered sender name (200,000 VND once, then 200,000 VND a month, about \$7.70). Approval takes 3–5 working days.
eSMS, Stringee, VNPT, Viettel (local)	price not shown on-line	Same kind of service; ask for a written quote.
Zalo ZNS	about \$0.008 (200 VND)	Cheap, but only reaches people who use Zalo. Could be the first channel with text messages as backup.
Plivo	\$0.069–0.228	Cheapest route is a small carrier; Viettel, the biggest, costs \$0.193. Blended about \$0.157.
Bird (MessageBird)	about \$0.155	Built into Supabase.
Twilio	\$0.2852	One flat rate, built into Supabase.
Firebase	not listed for Vietnam	\$0.01 in the US; needs a billing account.

United States (phone numbers starting +1)

Provider	Price per text	What to know
Telnyx	about \$0.008	Lowest price found. Plugs into Supabase through the “send SMS hook”.
Plivo, Sinch, MSG91	\$0.0065–0.013	Low prices, no per-verification fee.
Firebase	\$0.01	A few free texts a day.
Twilio (plain text)	about \$0.013	Built into Supabase; a safe default.
AWS	about \$0.02	No fraud protection included.
Twilio Verify	\$0.05 + text (about \$0.07)	Includes fraud protection, but the fee is about six times the text itself.
Vonage Verify	about \$0.07	Similar to Twilio Verify.

What one city (5,000 users) pays each month for verification texts



All four stages, per month	A	B	C	D
Vietnam: local at 450 VND	\$13	\$47	\$163	\$525
Vietnam: local at 590 VND (used above)	\$15	\$59	\$211	\$686
Vietnam: local at 790 VND	\$17	\$76	\$280	\$916
Vietnam: Plivo	\$47	\$353	\$1,413	\$4,710
Vietnam: Twilio	\$86	\$642	\$2,567	\$8,556
US: Telnyx	\$2	\$18	\$72	\$240
US: Twilio text	\$4	\$29	\$117	\$390
US: Twilio Verify	\$14	\$104	\$417	\$1,390
Mixed 70% Vietnam / 30% US, cheapest pair	\$11	\$46	\$169	\$552
Mixed 70% / 30%, Twilio for both	\$61	\$458	\$1,832	\$6,106

Fraud scenario: fake sign-ups. Bots can request thousands of codes to run up your bill (“SMS pumping”). If 5,000 fake Vietnamese requests arrive in one day, the bill is about **\$1,426** on Twilio, **\$785** on Plivo, and about **\$113** on a Vietnamese local provider (a prepaid balance also caps the loss). The same attack against Twilio in the US costs about \$65. Set limits per phone number, per device and per country, and add a CAPTCHA, from day one.

Recommendation. For Vietnamese numbers use a local brandname provider (about 10–15 times cheaper than Twilio). For US numbers use Telnyx or Plivo, or plain Twilio because Supabase supports it directly. Avoid Twilio Verify unless you want its built-in fraud protection. Supabase supports Twilio, Twilio Verify, Bird, Vonage and Textlocal directly, and any other provider through its “send SMS hook” (a small piece of custom code).

7. What to do, in order

- ☐ **Upgrade Supabase to the Pro plan (\$25).** It fixes the pause immediately; it can be cancelled after a month.
- ☐ **Move the website to Vercel Pro (\$20)** before launching as a business (the free plan is for non-commercial projects only).
- ☐ **Choose the text-message providers** (one for Vietnam, one for the US) and get written prices. This is the biggest saving; section 6 compares them.
- ☐ **Replace Gmail for sending email** with a proper email service (free up to 3,000 emails a month).
- ☐ **Set up the photo wall (Cloudflare R2) before launch:** buy a web address, shrink the existing photos, then copy them over. One day of work, about \$12 a year.
- ☐ **Check the live phone-login setting** in the Supabase dashboard, raise the 30-texts-per-hour cap when needed, and turn on a CAPTCHA.
- ☐ **Check three screens each month:** Supabase usage, Cloudflare R2, and the text-message balance.

8. How sure are these numbers?

- Solid:** Plan prices for Supabase, Vercel, Cloudflare R2, Resend, Twilio, Plivo, Telnyx, SpeedSMS and Apple, read from each company's own page on 9 October 2026.
- Estimated:** How many people use the app, how many photos they view, how many emails and texts are sent, the share of texts going to each phone carrier, the prices of providers whose pages I could not read, and the maps and monitoring costs.
- Not included:** VAT and other taxes; payment fees paid by hosts (Vietnamese bank transfers, Stripe in the US); salaries, legal, and marketing; Vietnamese data-storage rules (ask a lawyer before choosing where the database sits).

Full detail, sources and assumptions: `docs/backend-cost-estimate.md`. Setup steps for Cloudflare R2: note 40 in `.claude/notes/`.